

Afondamento nas Competencias Profesionais

Práctica 1.5

CI/CD - Coolify y Ngrok

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1 Introducción

[Volver](#)

- ENUNCIADO -

- ENUNCIADO -

Requisitos previos:

- Creación de VM:
 - SO recomendado: Ubuntu.

2 Coolify, Ngrok

Volver

Instalación de Coolify y Ngrok:

<https://coolify.io/>

<https://ngrok.com/>

Tan solo hay que seguir las instrucciones en pantalla. Importante en Ngrok, crear cuenta y añadir Authtoken.

Self-hosted Installation

If you like taking control and managing everything yourself, self-hosting Coolify is the way to go.

It's completely free (apart from your server costs) and gives you full control over your setup.

⚡ Quick Installation (recommended):

```
curl -fsSL https://cdn.coollabs.io/coolify/install.sh | sudo bash
```

sh

Run this script in your terminal, and Coolify will be installed automatically. For more details, including firewall configuration and prerequisites, check out the guide below.

Note for Ubuntu Users:

The automatic installation script only works with Ubuntu LTS versions (20.04, 22.04, 24.04). If you're using a non-LTS version (e.g., 24.10), please use the [Manual Installation](#) method below.

Figura 1: Instalar Coolify

1

Connect

Get an endpoint online.



Agent

Linux

Choose another platform

Installation

[Apt](#) [Download](#) [Snap](#)

Install ngrok via Apt with the following command:

```
curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc \
| sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null \
&& echo "deb https://ngrok-agent.s3.amazonaws.com bookworm main" \
| sudo tee /etc/apt/sources.list.d/ngrok.list \
&& sudo apt update \
&& sudo apt install ngrok
```

Run the following command to add your authtoken to the default **ngrok.yml** configuration file.

```
ngrok config add-authtoken 36kqwDWy2W0EMplhxC15ba5wNff_7738oSN3TRkft4q7bEPz9
```

Figura 2: Instalar Ngrok

3 Prueba con API test simple

Volver

Ir al dashboard de Coolify (localhost:8000)

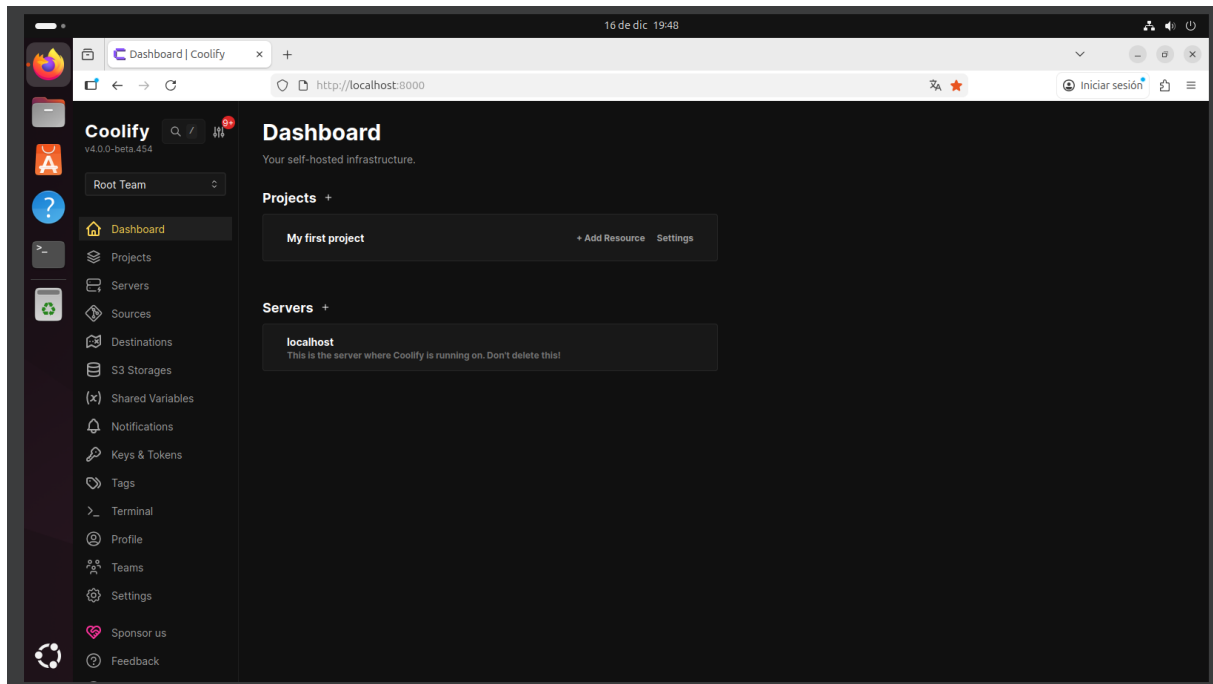


Figura 3: Dashboard

Crear nuevo proyecto, añadir recurso (repositorio público, <https://github.com/1001inchNails/apiTestCicd>)

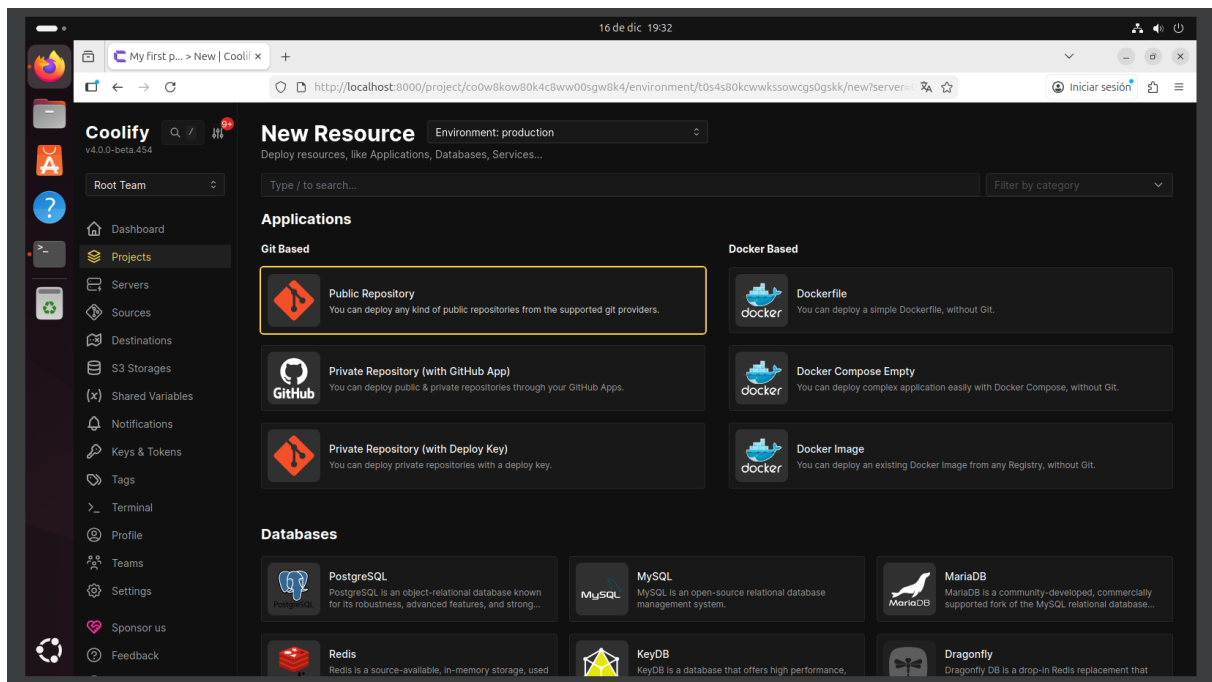


Figura 4: Añadir recurso

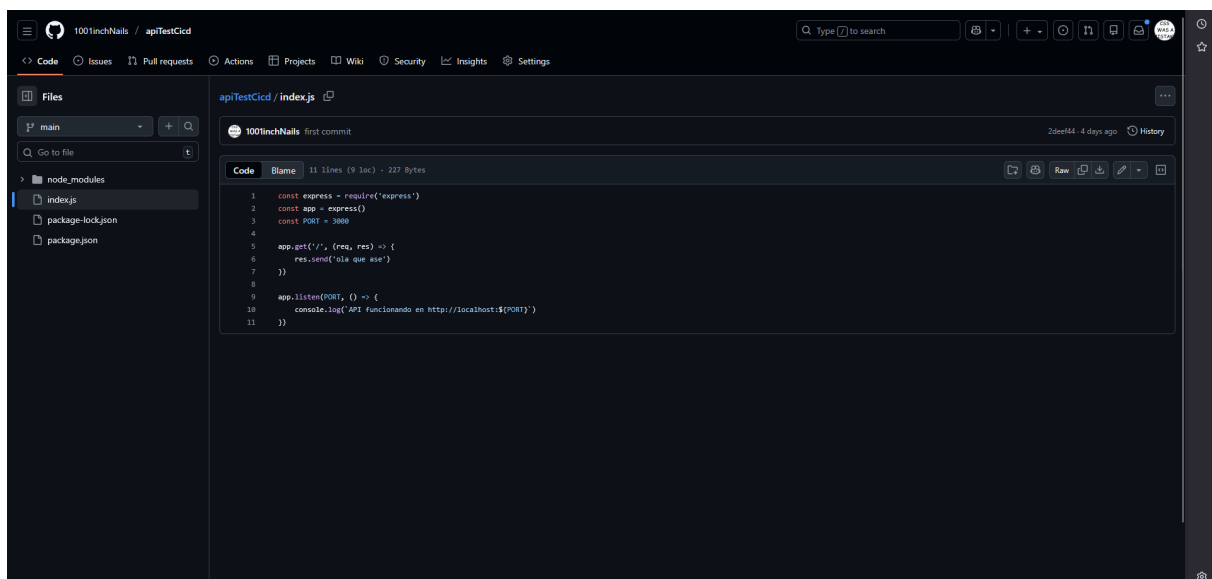


Figura 5: Test API

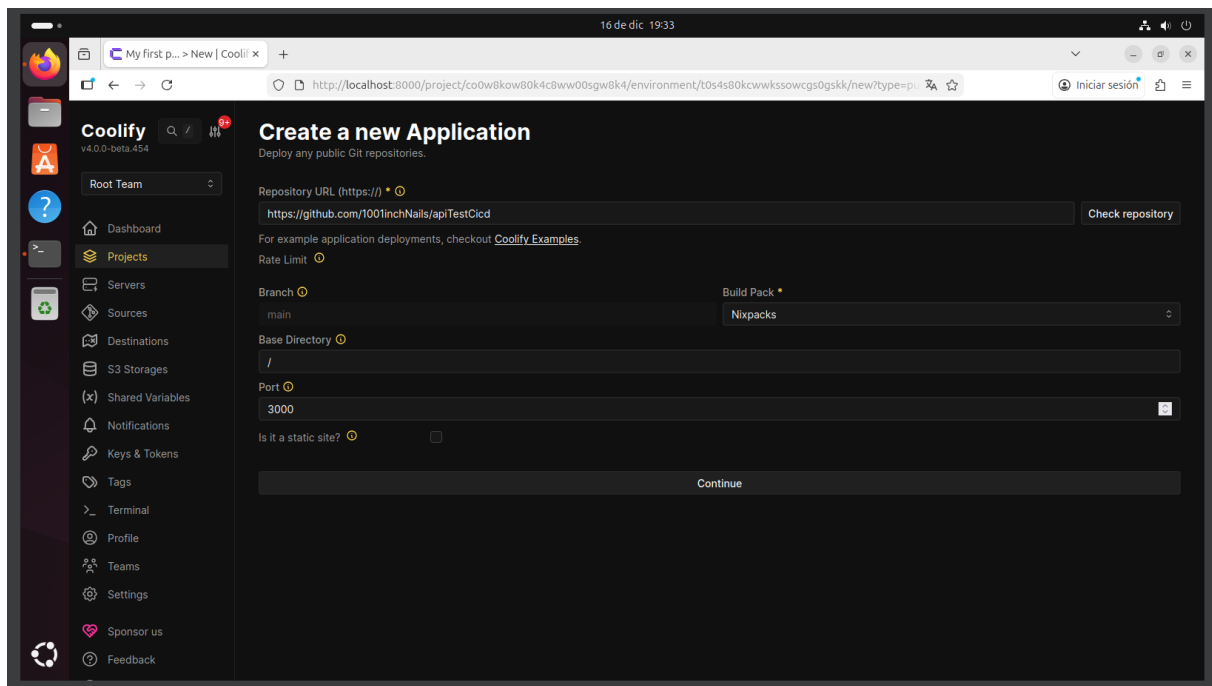


Figura 6: Nixpacks, puerto 3000

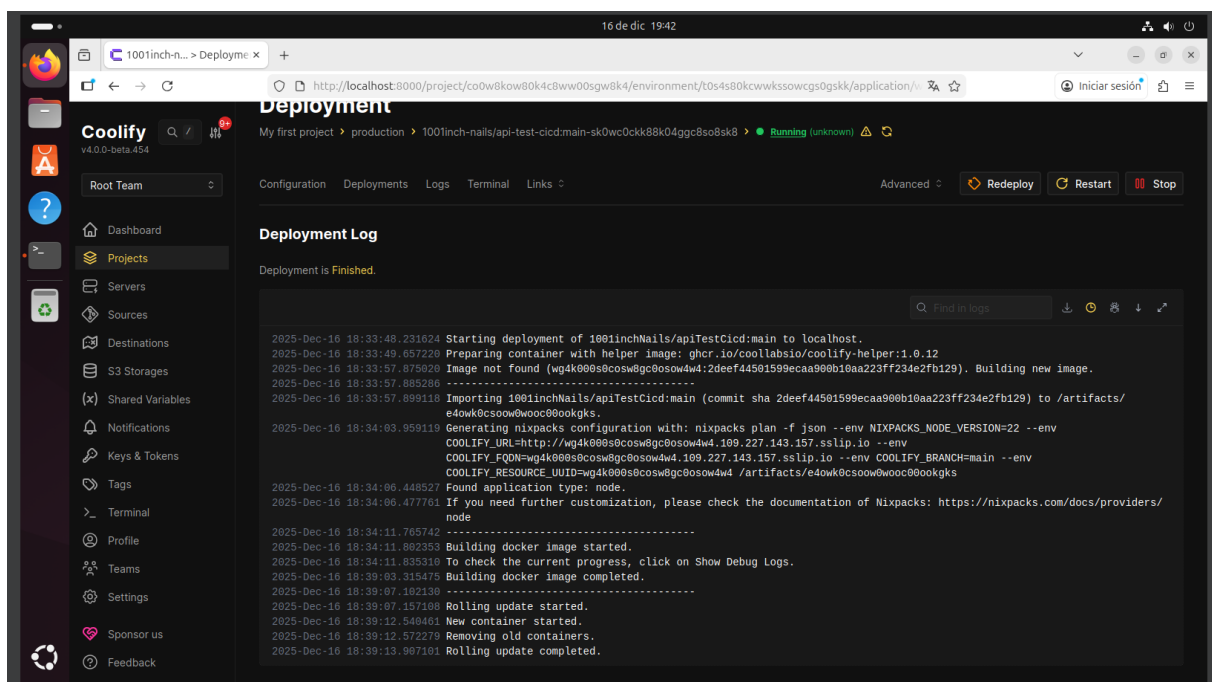


Figura 7: Desplegar la API

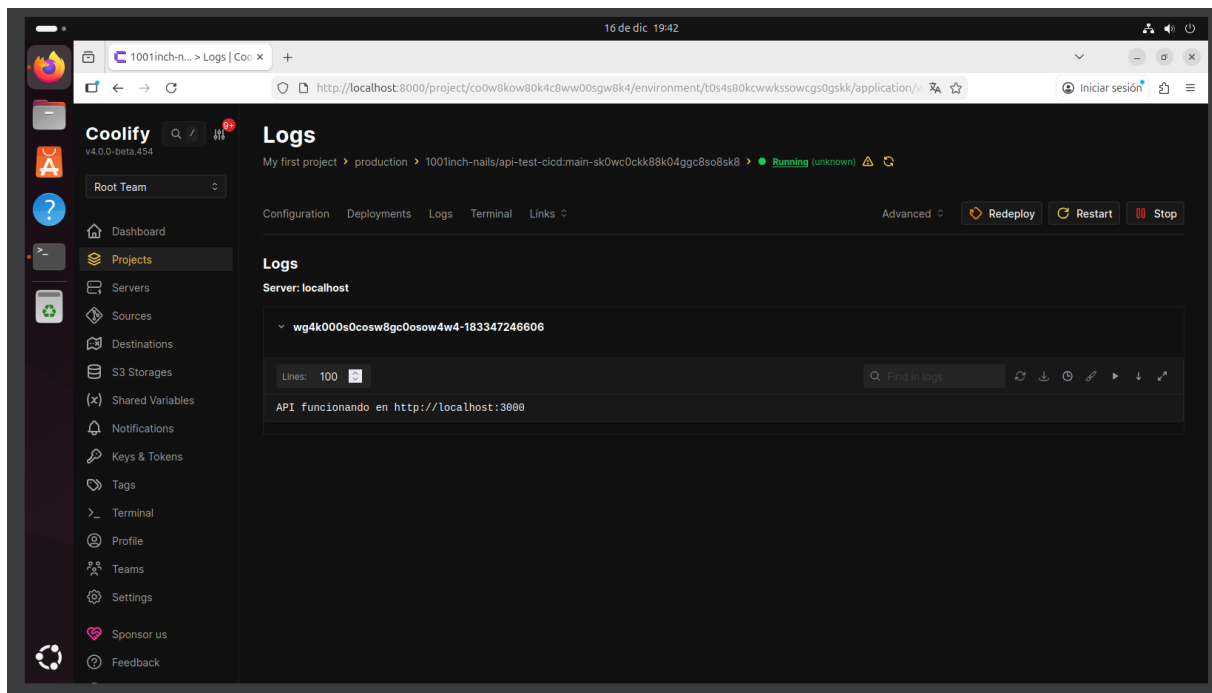


Figura 8: Comprobar logs

Buscamos la ID del contenedor con nuestra API:

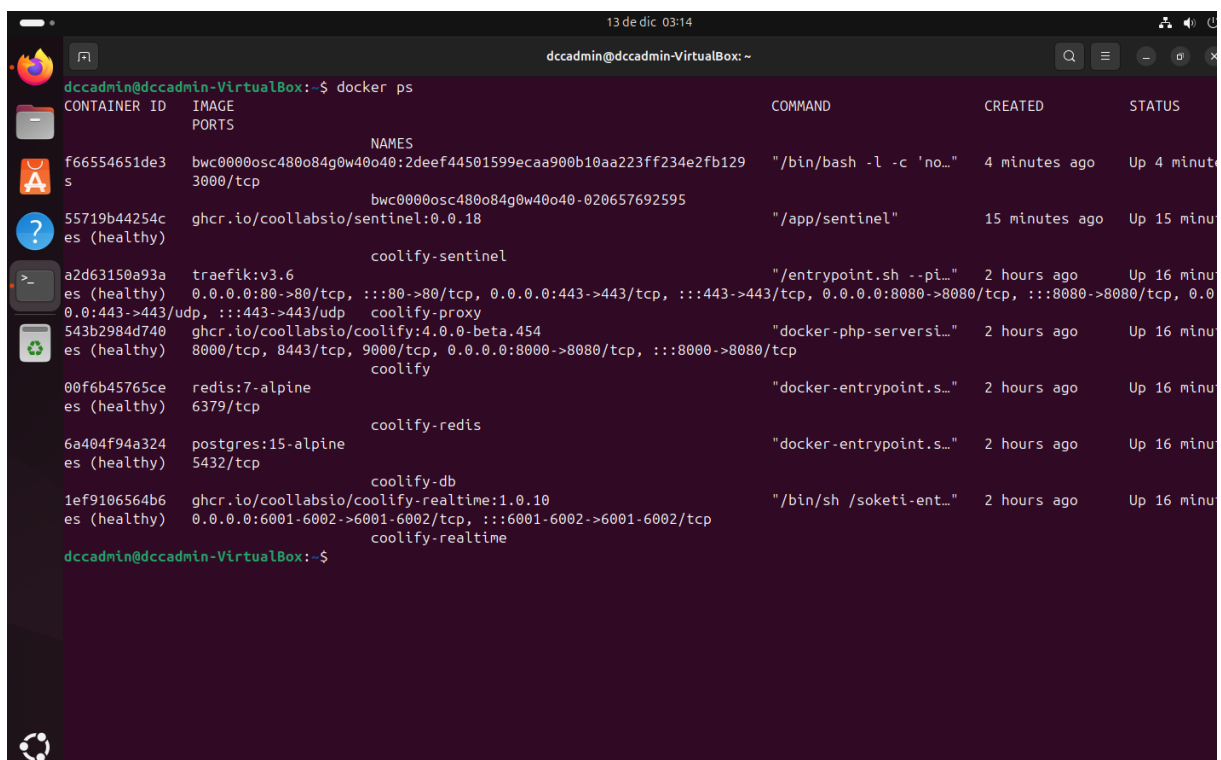


Figura 9: En este caso, el primero

Con esa ID, buscamos la IP del contenedor:

```
1ef9106564b6 ghcr.io/coolabsio/coolify-realtime:1.0.10 "/bin/sh /soketi-ent..." 2 hours ago Up 16 min
es (healthy) 0.0.0.0:6001-6002->6001-6002/tcp, :::6001-6002->6001-6002/tcp
coolify-realtime
dccadmin@dccadmin-VirtualBox:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' f66554651de3
10.0.1.8
dccadmin@dccadmin-VirtualBox:~$
```

Figura 10: IP del contenedor con nuestra API

Comprobación en máquina local:



Figura 11: Comprobación

Exponemos con Ngrok:

```
dccadmin@dccadmin-VirtualBox:~$ docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' f66554651de3
10.0.1.8
dccadmin@dccadmin-VirtualBox:~$ ngrok http 10.0.1.8:3000
```

Figura 12: Creación de "túnel" para exponer la API de local a exterior

```
ngrok
Call internal services from your gateway: https://ngrok.com/r/http-request

Session Status      online
Account             danielcc@aulaestudio.es (Plan: Free)
Version             3.34.1
Region              Europe (eu)
Web Interface        http://127.0.0.1:4040
Forwarding            https://magdalene-parotidean-deceptively.ngrok-free.dev -> http://10.0.1.8:3000

Connections
ttl    opn    rt1    rt5    p50    p90
0       0       0.00   0.00   0.00   0.00
```

Figura 13: No cerrar el terminal

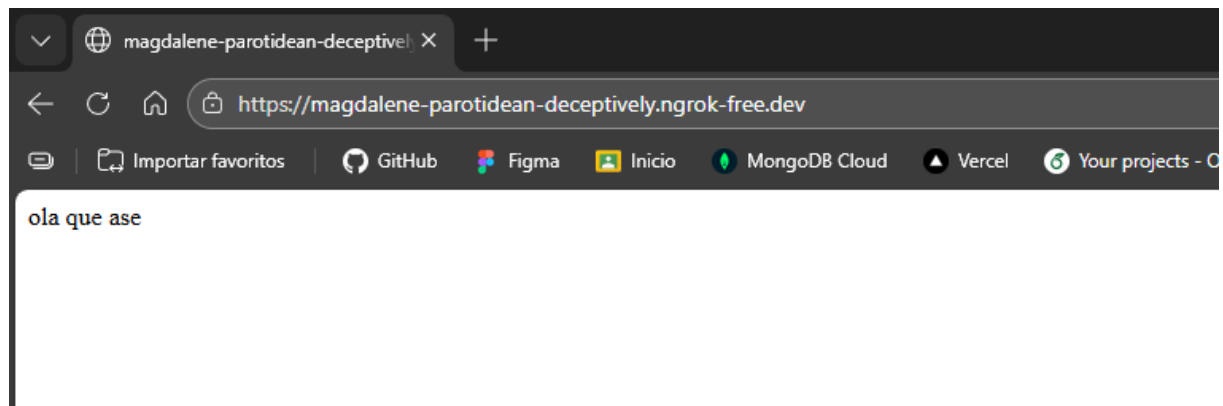


Figura 14: Prueba en maquina host

4 Prueba con API/BBDD

Volver

La práctica se desarrollará desde la máquina host, para ello exponemos el puerto 8000, usado por el dashboard de Coolify:

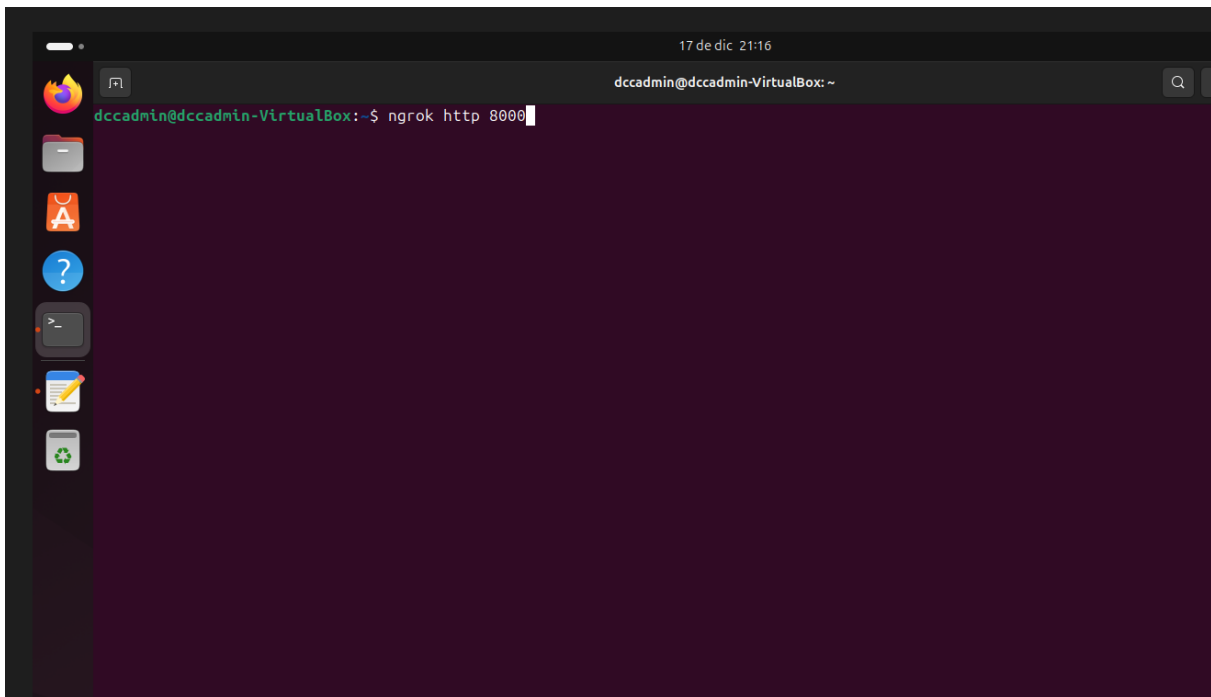


Figura 15: Exponer puerto

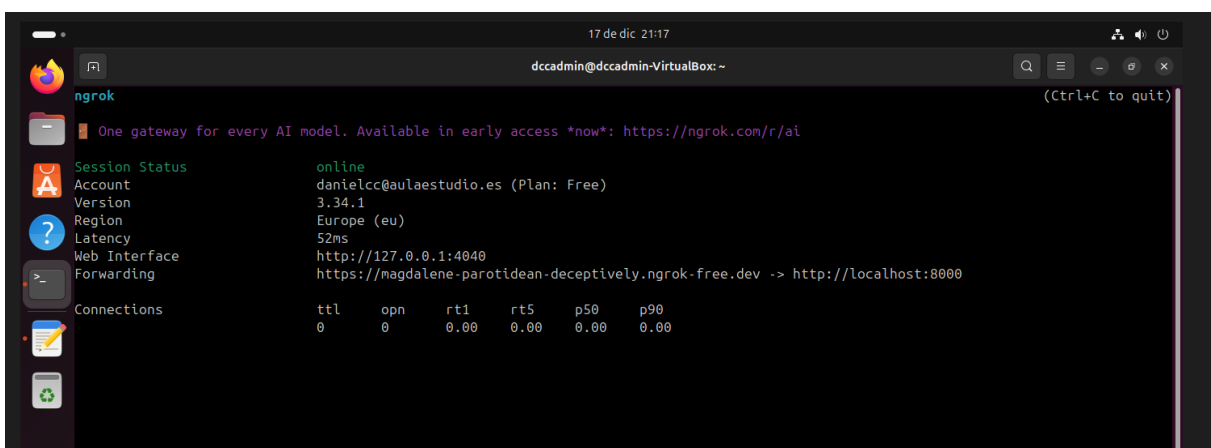


Figura 16: Tunnel

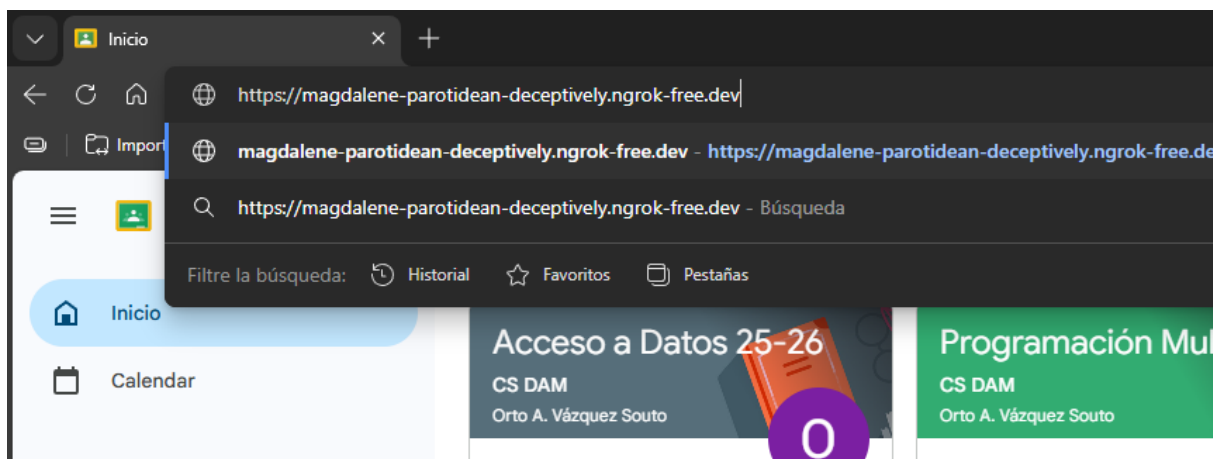


Figura 17: En host

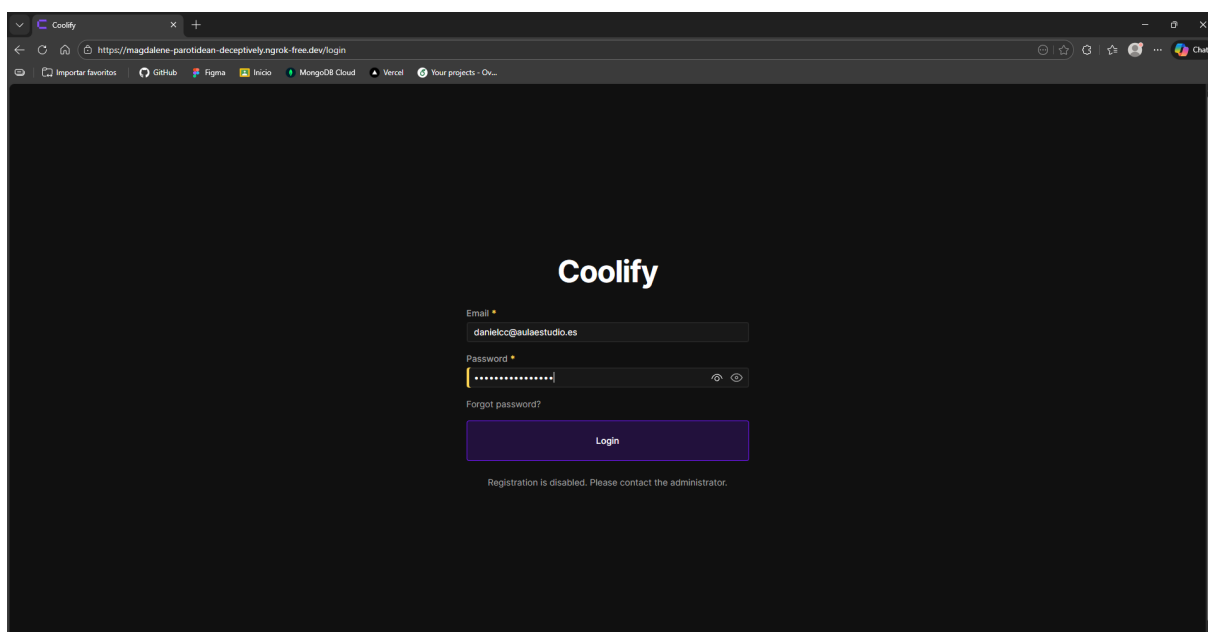


Figura 18: Acceso a dashboard

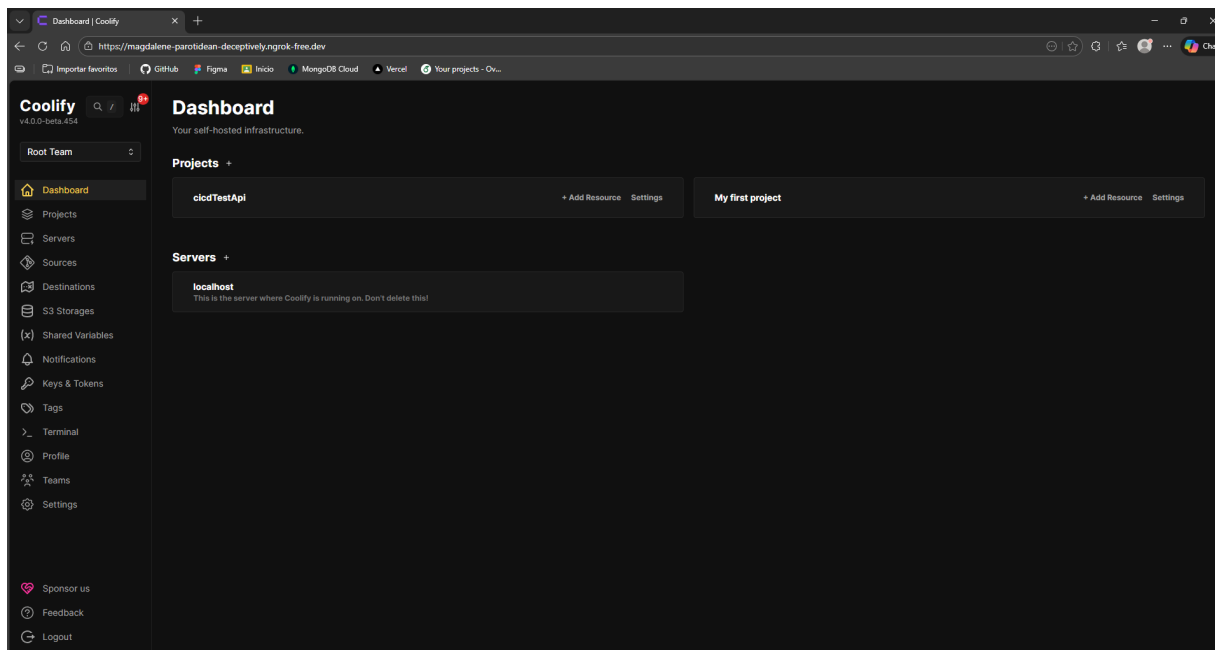


Figura 19: Dashboard de Coolify de VM en máquina host

Creamos un nuevo proyecto:

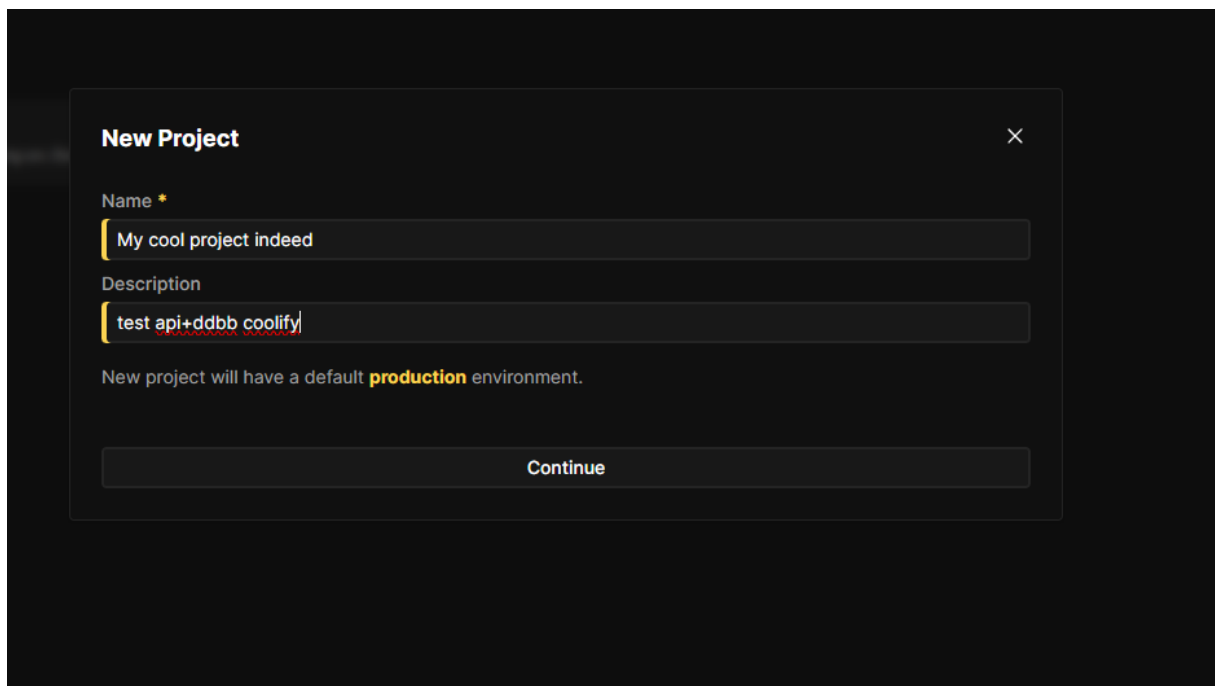


Figura 20: Crear nuevo proyecto

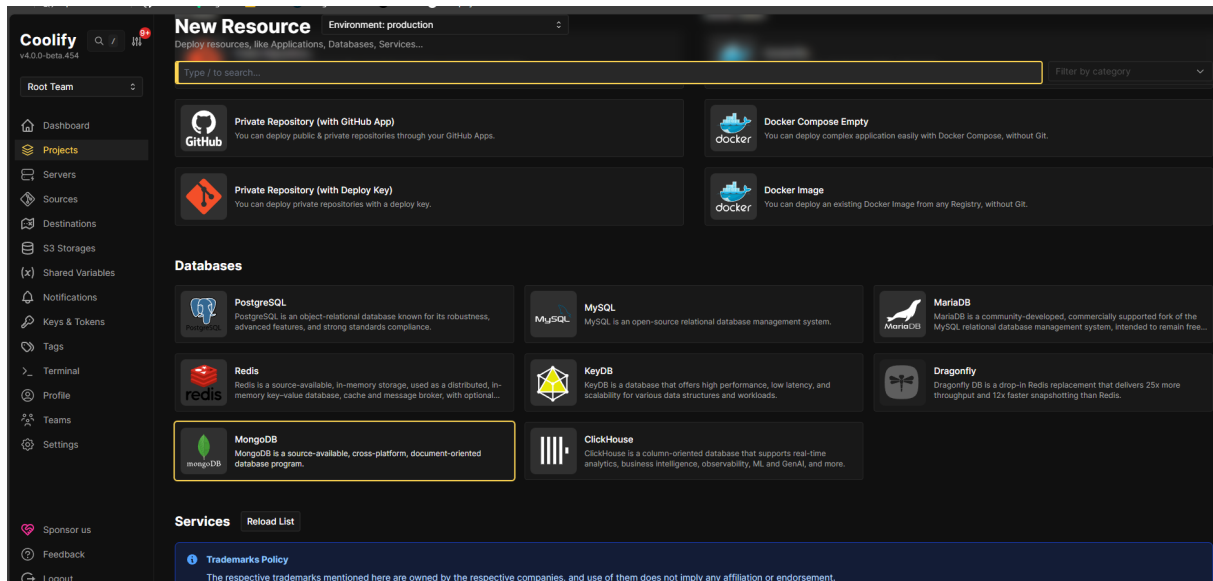


Figura 21: Añadimos recurso: BBDD

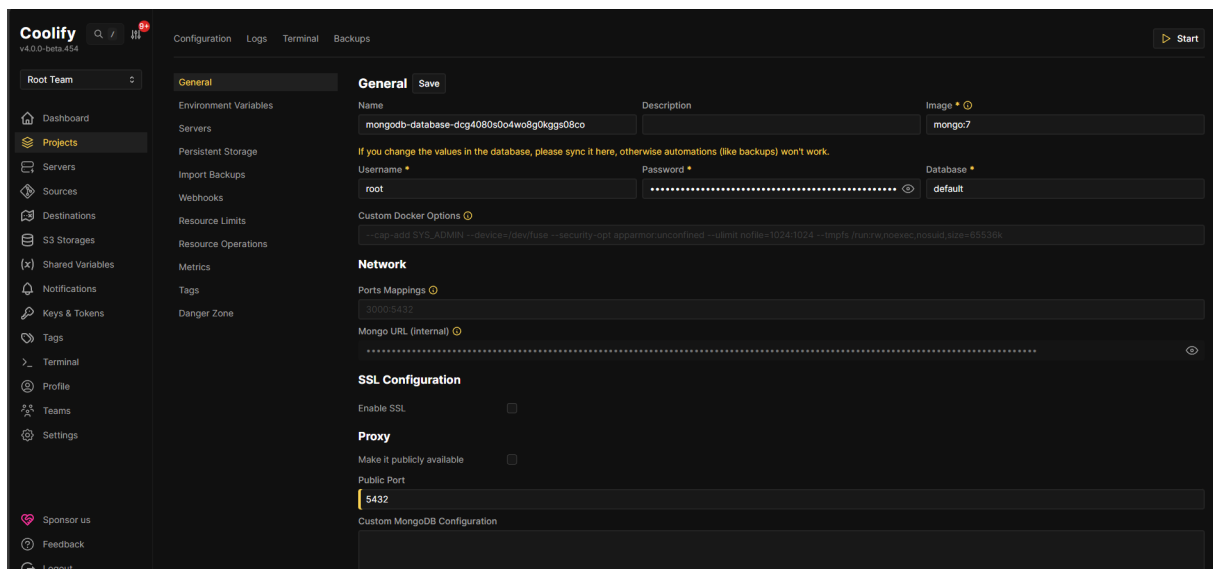


Figura 22: Configurar puertos

Ponerle la contraseña generada en la variable de entorno:

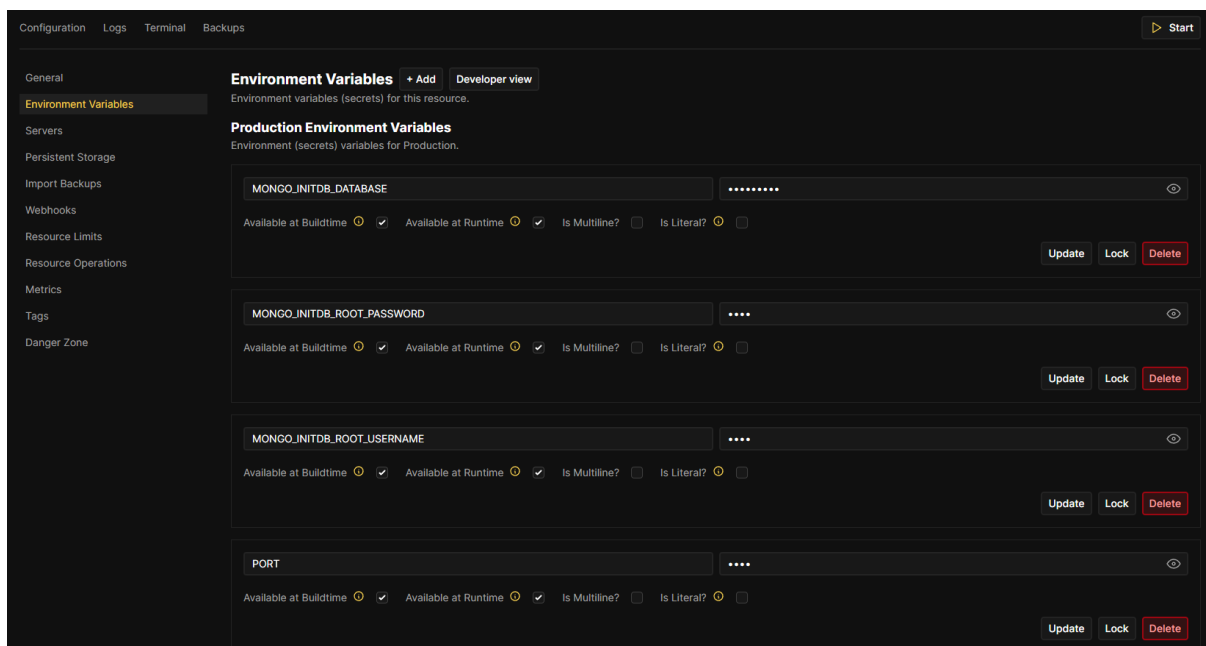


Figura 23: Meter variables de entorno

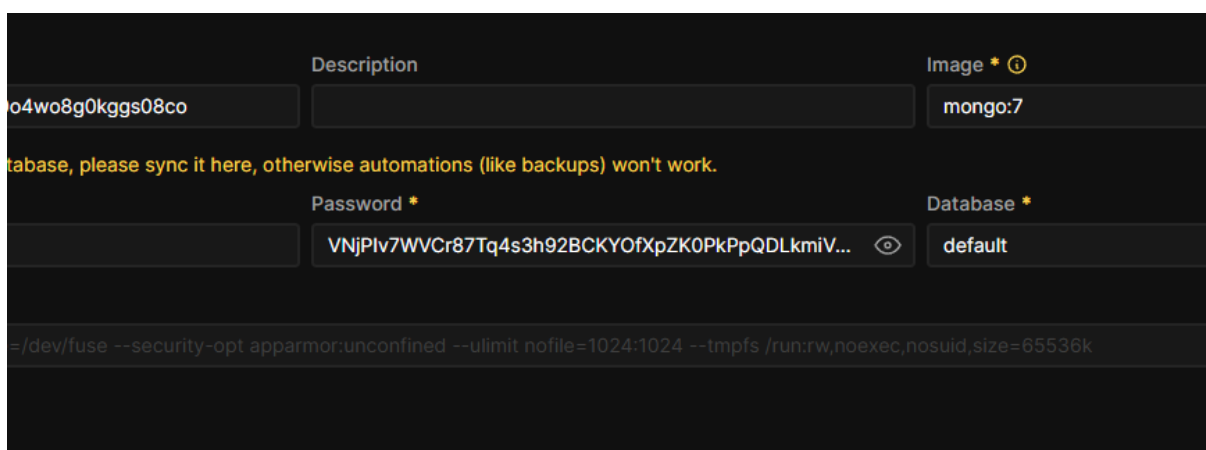


Figura 24: Esta

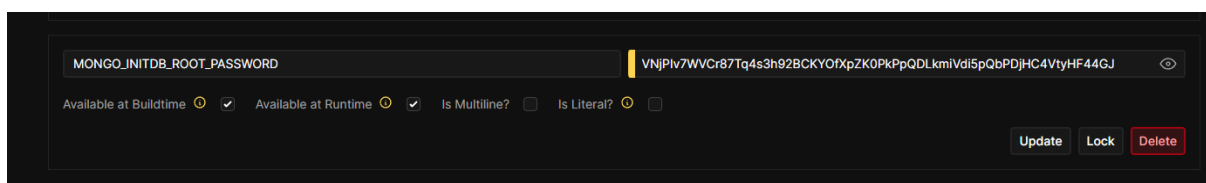
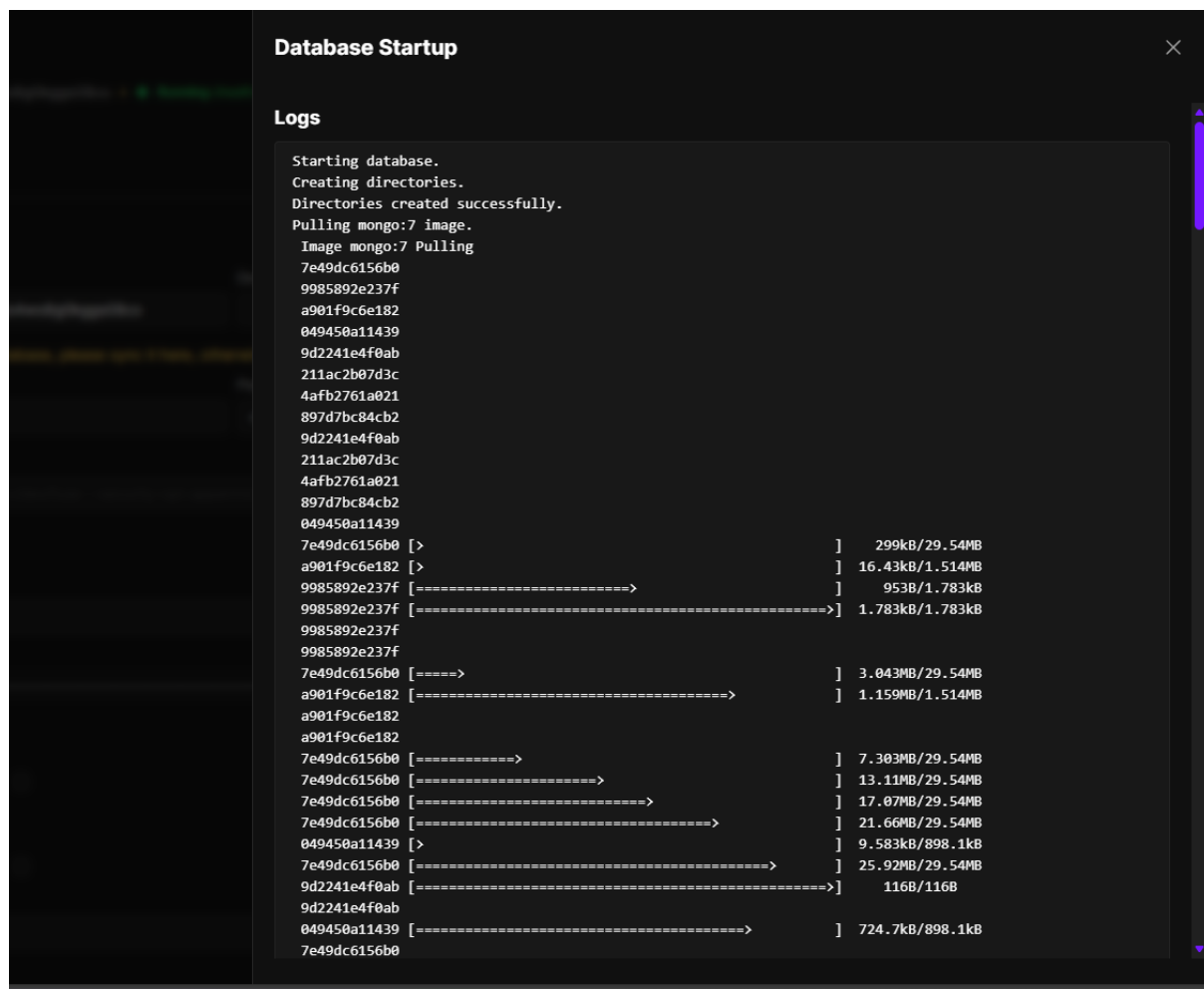


Figura 25: Aquí



Database Startup

Logs

```
Starting database.
Creating directories.
Directories created successfully.
Pulling mongo:7 image.
Image mongo:7 Pulling
7e49dc6156b0
9985892e237f
a901f9c6e182
049450a11439
9d2241e4f0ab
211ac2b07d3c
4afb2761a021
897d7bc84cb2
9d2241e4f0ab
211ac2b07d3c
4afb2761a021
897d7bc84cb2
049450a11439
7e49dc6156b0 [ > ] 299kB/29.54MB
a901f9c6e182 [ > ] 16.43kB/1.514MB
9985892e237f [=====>] 953B/1.783kB
9985892e237f [=====>] 1.783kB/1.783kB
9985892e237f
9985892e237f
7e49dc6156b0 [=====>] 3.043MB/29.54MB
a901f9c6e182 [=====>] 1.159MB/1.514MB
a901f9c6e182
a901f9c6e182
7e49dc6156b0 [=====>] 7.303MB/29.54MB
7e49dc6156b0 [=====>] 13.11MB/29.54MB
7e49dc6156b0 [=====>] 17.07MB/29.54MB
7e49dc6156b0 [=====>] 21.66MB/29.54MB
049450a11439 [ > ] 9.583kB/898.1kB
7e49dc6156b0 [=====>] 25.92MB/29.54MB
9d2241e4f0ab [=====>] 116B/116B
9d2241e4f0ab
049450a11439 [=====>] 724.7kB/898.1kB
7e49dc6156b0
```

Figura 26: Desplegamos

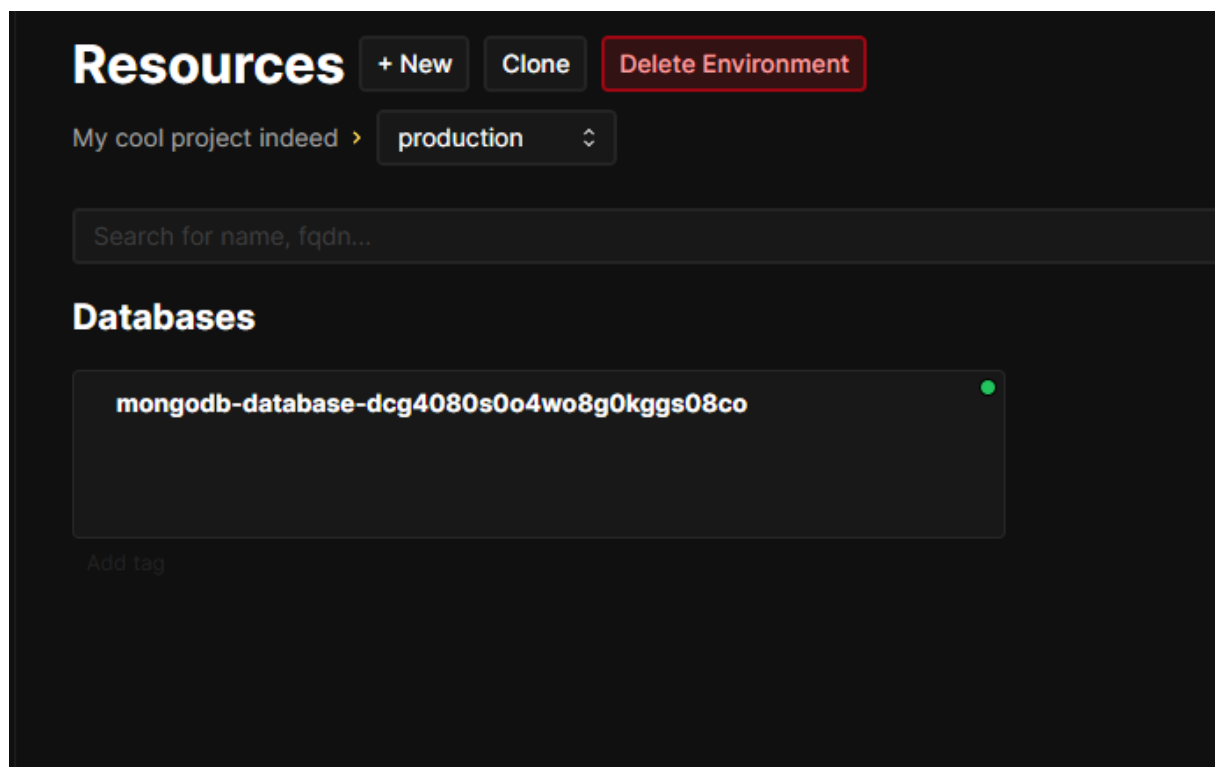


Figura 27: Despliegue correcto

Ahora creamos nuevo recurso, en este caso con la API creada en la práctica anterior (https://github.com/1001inchNails/afond_1-4).

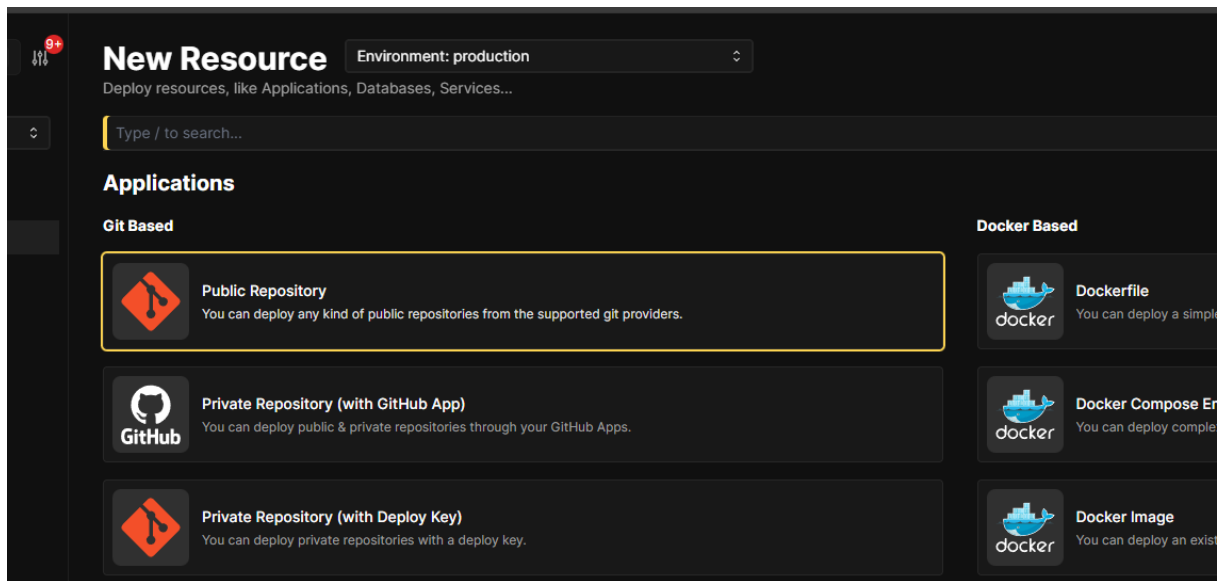


Figura 28: Crear nuevo recurso

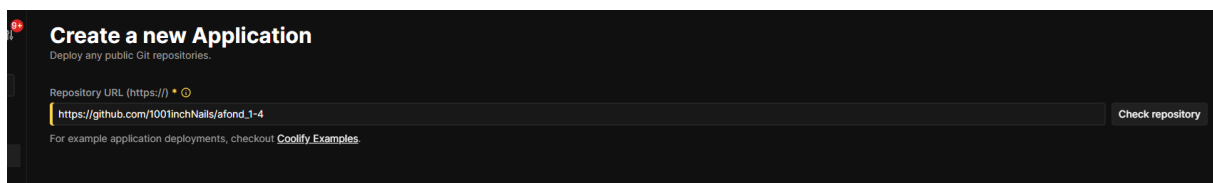


Figura 29: Repositorio público

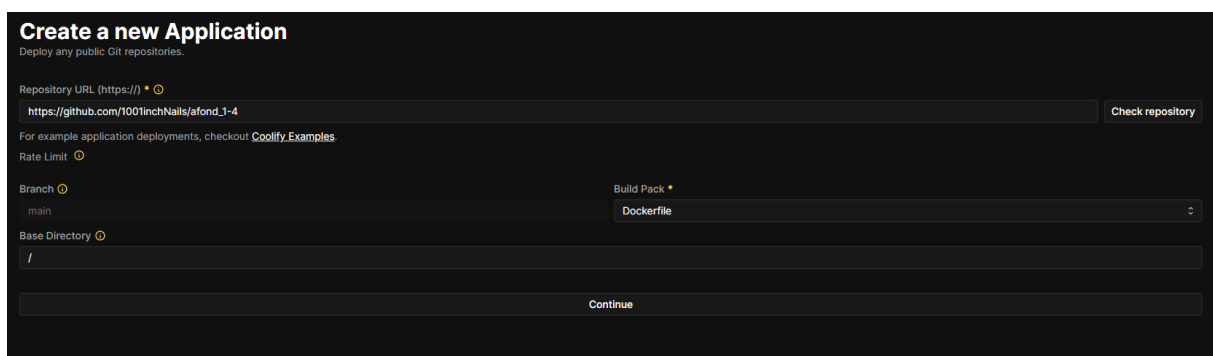


Figura 30: Build: Dockerfile

Environment Variables + Add Developer view

Environment variables (secrets) for this resource.

Sort alphabetically

Use Docker Build Secrets

Production Environment Variables

Environment (secrets) variables for Production.

MONGO_URI [password mask]

Available at Buildtime ☒ Available at Runtime ☒ Is Multiline? ☐ Is Literal? ☐

Update Lock Delete

PORT [text input]

Available at Buildtime ☒ Available at Runtime ☒ Is Multiline? ☐ Is Literal? ☐

Update Lock Delete

Preview Deployments Environment Variables

Figura 31: Variables de entorno: meterle el password en MONGO_URI

General Save

General configuration for your application.

Name * 1001inch-nails/afond_1-4:main-vs8w8cg4owgcsw40ccwoco8 Description

Build Pack * Dockerfile

Domains http://x4skssk0wk4wgcscckwwgcwc.91.117.232.105.sslip.io Generate Domain

Direction * Allow www & non-www. Set Direction

Docker Registry

Docker Image Empty means it won't push the image to a docker registry. Docker Image Tag Empty means only push commit sha tag.

Build

Base Directory / Dockerfile Location /Dockerfile Docker Build Stage Target

Custom Docker Options --cap-add SYS_ADMIN --device=/dev/fuse --security-opt apparmor:unconfined --ulimit nofile=1024:1024 --tmpfs /run,runw,nosuid,size=65536k --hostname=myapp

Use a Build Server? ☐

Figura 32: Definir carpeta raiz, y ponerle el mismo puerto que el definido en el código de la API

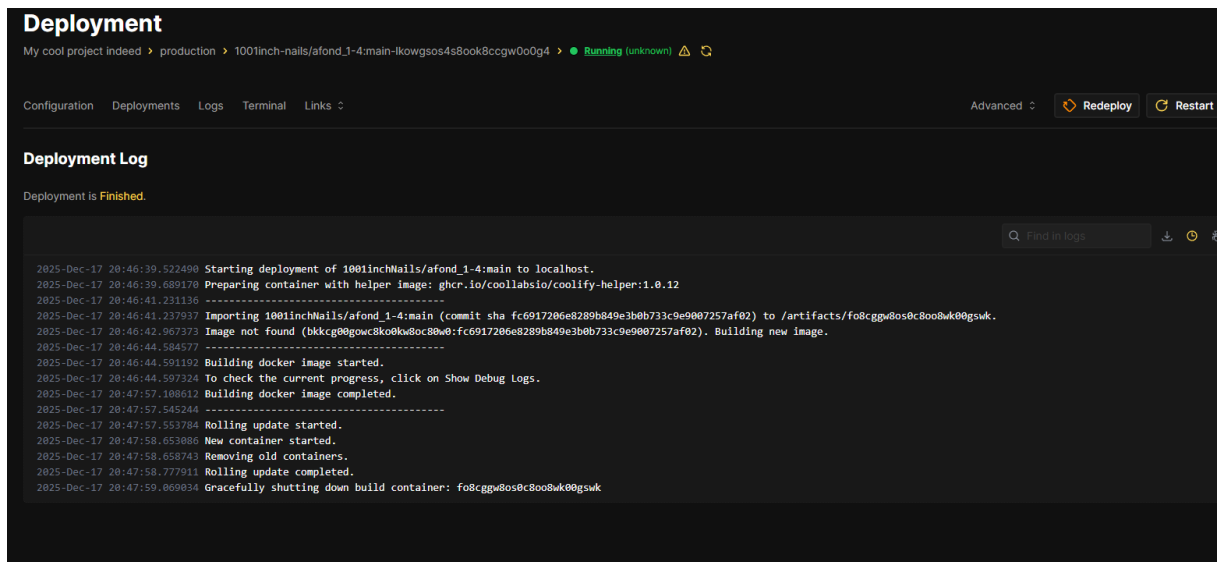


Figura 33: Guardar y desplegar

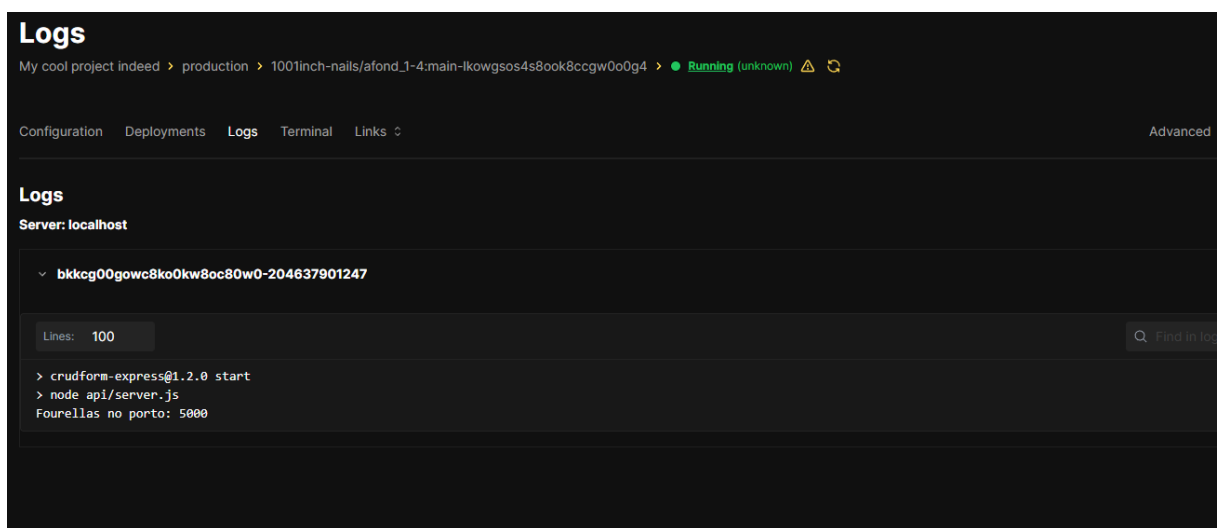
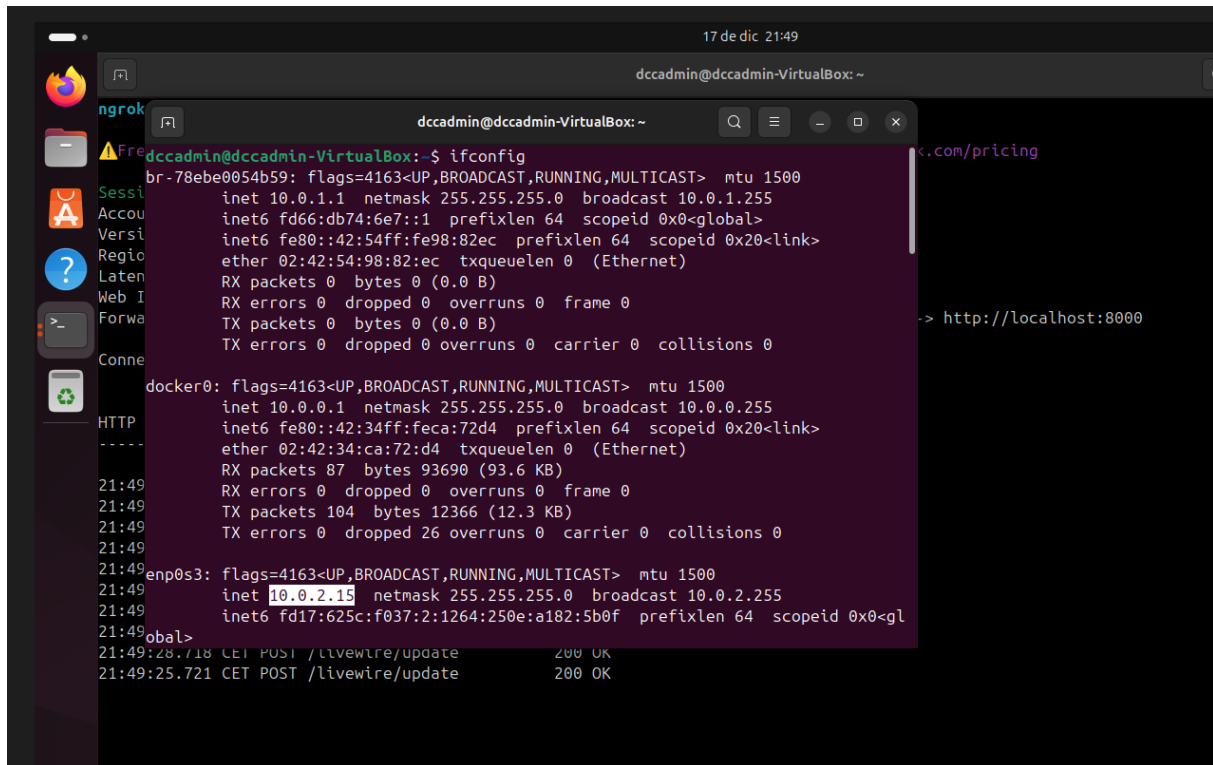


Figura 34: Log

Ahora buscamos la IP de la VM:



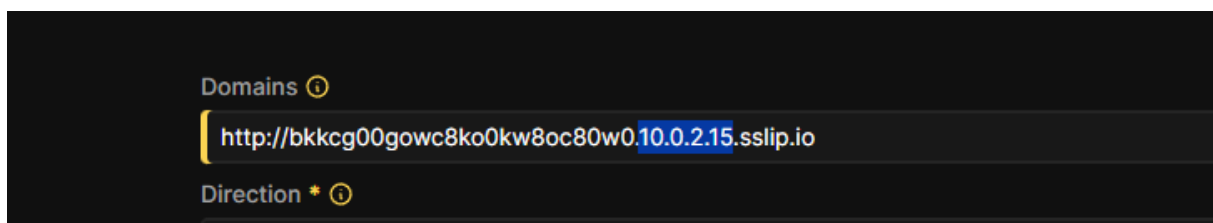
```
17 de dic 21:49
dccadmin@dccadmin-VirtualBox: ~
$ ifconfig
br-78be0054b59: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.1.1 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fd66:db74:6e7::1 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::42:54ff:fe98:82ec prefixlen 64 scopeid 0x20<link>
    ether 02:42:54:98:82:ec txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

docker0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.255.255.0 broadcast 10.0.0.255
    inet6 fe80::42:34ff:feca:72d4 prefixlen 64 scopeid 0x20<link>
    ether 02:42:34:ca:72:d4 txqueuelen 0 (Ethernet)
    RX packets 87 bytes 93690 (93.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 104 bytes 12366 (12.3 KB)
    TX errors 0 dropped 26 overruns 0 carrier 0 collisions 0

enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:1264:250e:a182:5b0f prefixlen 64 scopeid 0x0<gl
    global>
21:49:28.718 CET POST /livewire/update 200 OK
21:49:25.721 CET POST /livewire/update 200 OK
```

Figura 35: Interfaces de red (enp0s3)

Metemos esa IP en el dominio que nos proporciona Coolify:



```
Domains ⓘ
http://bkkcg00gowc8ko0kw8oc80w0.10.0.2.15.sslip.io
Direction * ⓘ
```

Figura 36: sslip

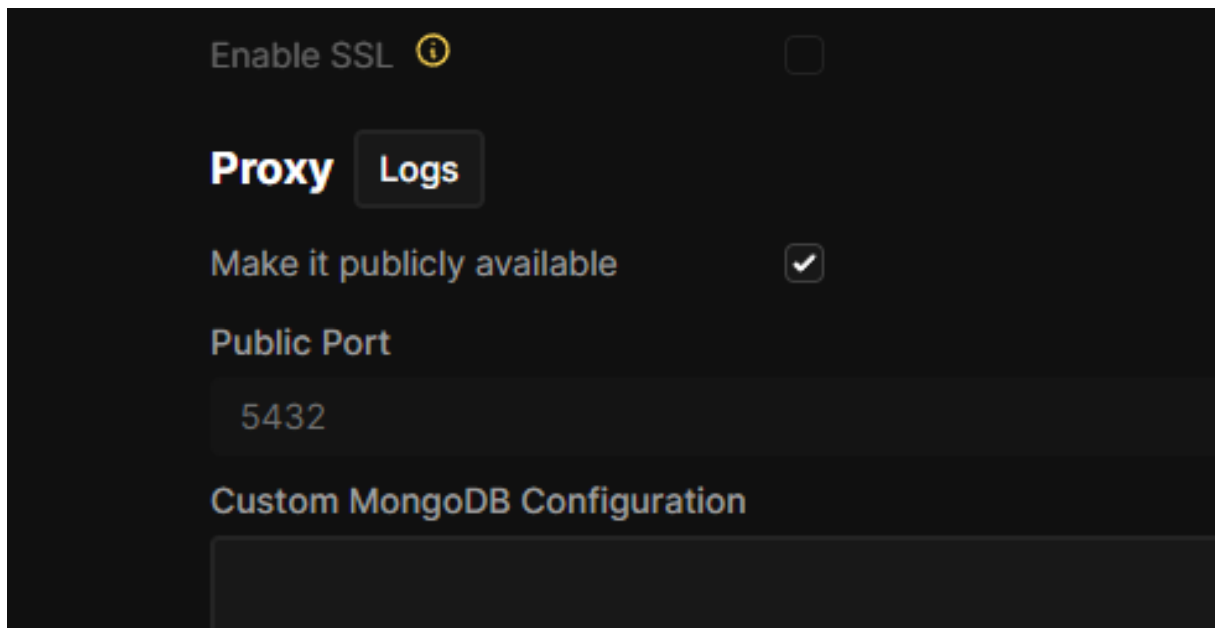


Figura 37: Hacerlo disponible a público

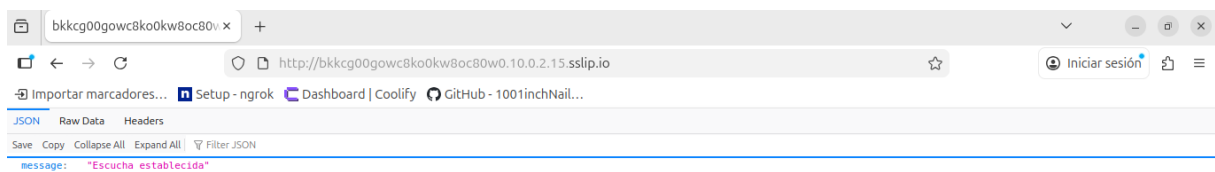


Figura 38: Correcto funcionamiento: VM

En este momento se descubrió un problema, en la VM el enlace cargaba correctamente pero no en la máquina host. Esto se debía a que la configuración de red de la VM no estaba en modo puente.

Después de cambiarlo, se procedió a volver a ejecutar `ifconfig` para buscar la IP nueva, meterla en el dominio y, esta vez, cargaba correctamente en host:

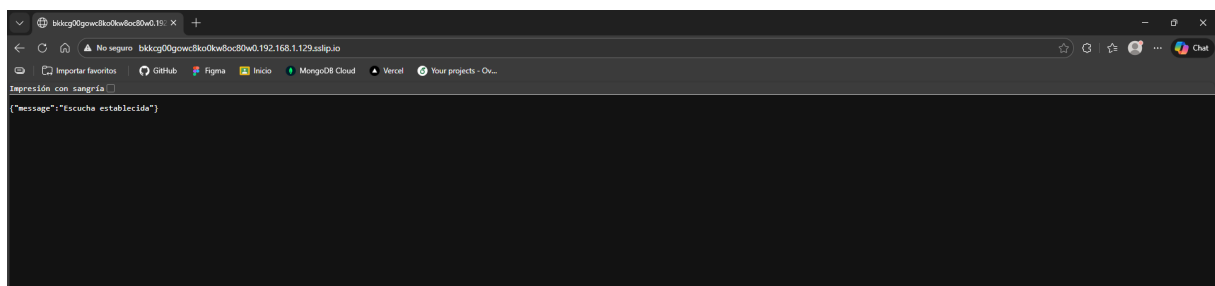


Figura 39: MISSION ACCOMPLISHED

5 Abrir tunel fuera de la red

Volver

Para abrir un tunel con ngrok y poder ver la API fuera de nuestra red, necesitamos usar este comando, usando el puerto 80 (por defecto de ngrok) y usar el dominio del webhook proporcionado por coolify:

```
ngrok http 80 --host-header=l0wcs0g0wcscg0occc8gg8cw.192.168.0.66.sslip.io
```

Con el enlace resultante podemos ver la API en cualquier dispositivo.

6 Auto-desplegar con webhook de Coolify

Volver

Para que la aplicación se auto despliegue cada vez que se hace un push al repositorio de github de la API.

Activar auto deploy en la configuración avanzada del proyecto (Coolify dashboard)

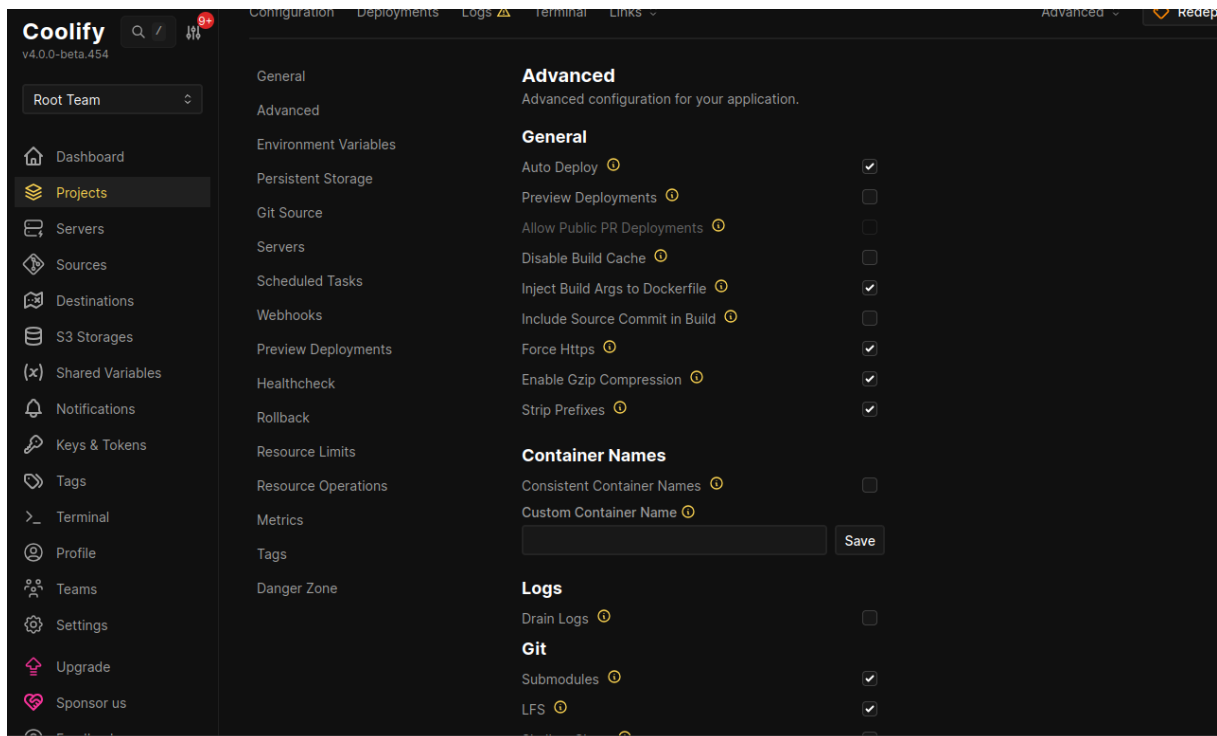


Figura 40: Activar auto deploy

Ir a la pestaña de Webhooks, crear un secret y ponerselo.

Webhooks ⓘ

Deploy Webhook (auth required) ⓘ

`http://109.227.134.36:8000/api/v1/deploy?uuid=l0wscg0wscsg0occc8gg8cw&force=false`

Manual Git Webhooks

GitHub ⓘ

`http://109.227.134.36:8000/webhooks/source/github/events/manual`

Webhook Configuration on GitHub ↗

GitHub Webhook Secret ⓘ

`6eaa9fafd9397f374d57ae91e971b2a1` ⓘ

GitLab

`http://109.227.134.36:8000/webhooks/source/gitlab/events/manual`

GitLab Webhook Secret ⓘ

Bitbucket

`http://109.227.134.36:8000/webhooks/source/bitbucket/events/man`

Bitbucket Webhook Secret ⓘ

Gitea

`http://109.227.134.36:8000/webhooks/source/gitea/events/manual`

Gitea Webhook Secret ⓘ

Save

Figura 41: Introducir secret

Ir a la pestaña Settings del repositorio de nuestra API, clickea Webhooks.

En payload URL introduce el Manual Git Webhook de Github, pero debes sustituir el dominio anterior a /webhooks por el dominio que nos da ngrok después de introducir el comando `ngrok http 8000`.

Ponemos el mismo secret que pusimos en la configuración de Coolify

Comprobamos que haga el ping correctamente

Webhooks Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

✓ `https://magdalene-parotidean-dece... (push)` Edit Delete

Last delivery was successful.

Figura 42: Ver ping

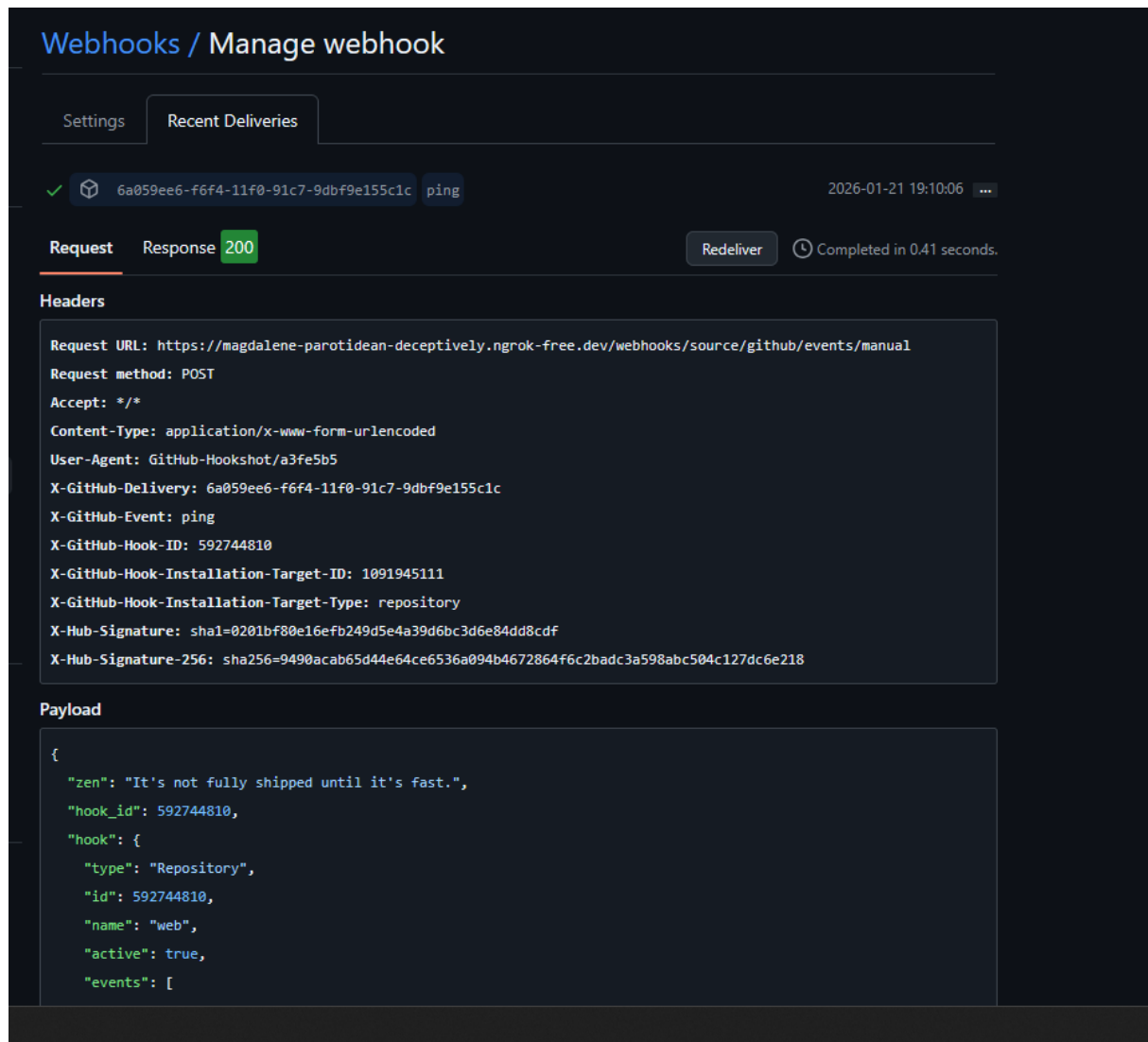


Figura 43: Success!

Ahora comprobamos con el webhook de Coolify <http://10wcs0g0wscsg0occc8gg8cw.192.168.0.68.sslip.io> el estado actual de la API

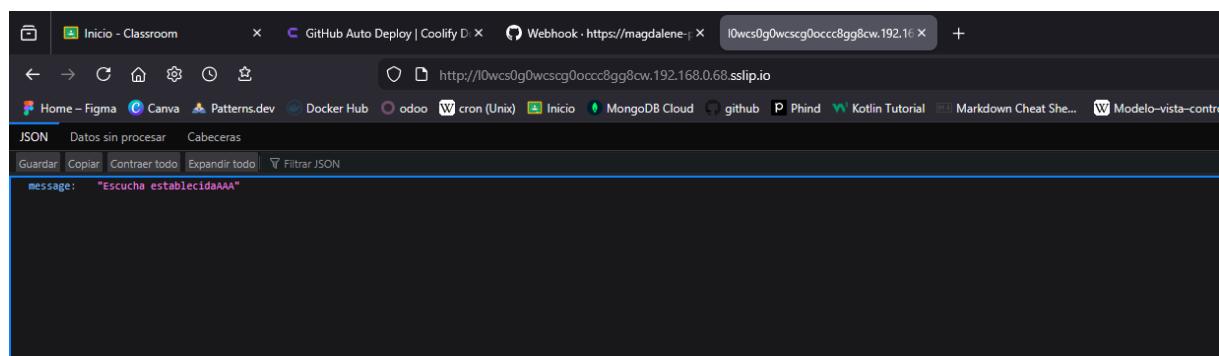


Figura 44: Estado actual

Hacemos un cambio menor a nuestra API.

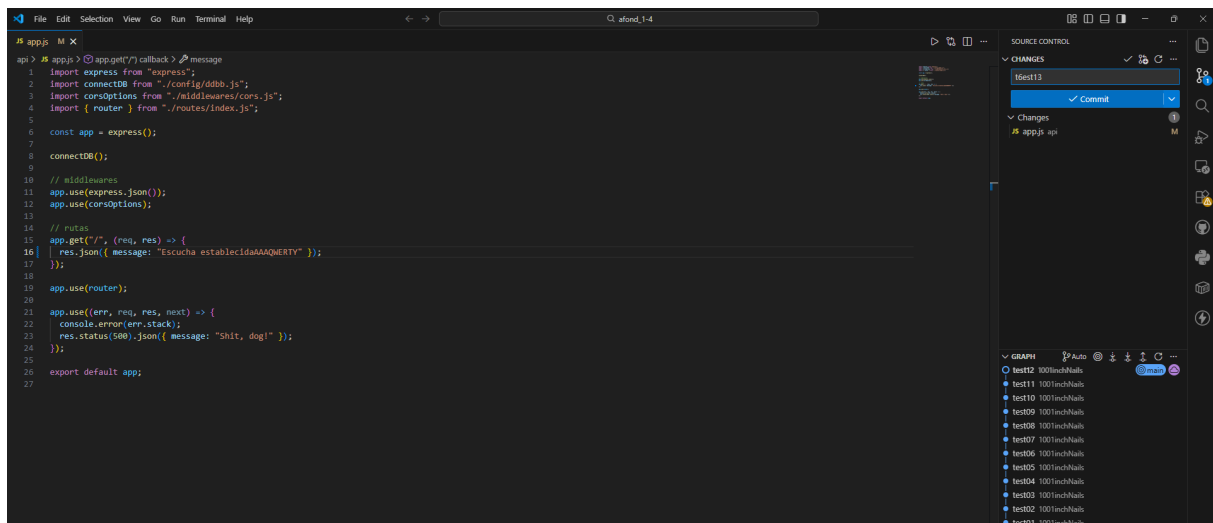


Figura 45: Cambio menor QWERTY

Comprobamos en Github que el push se haya detectado correctamente

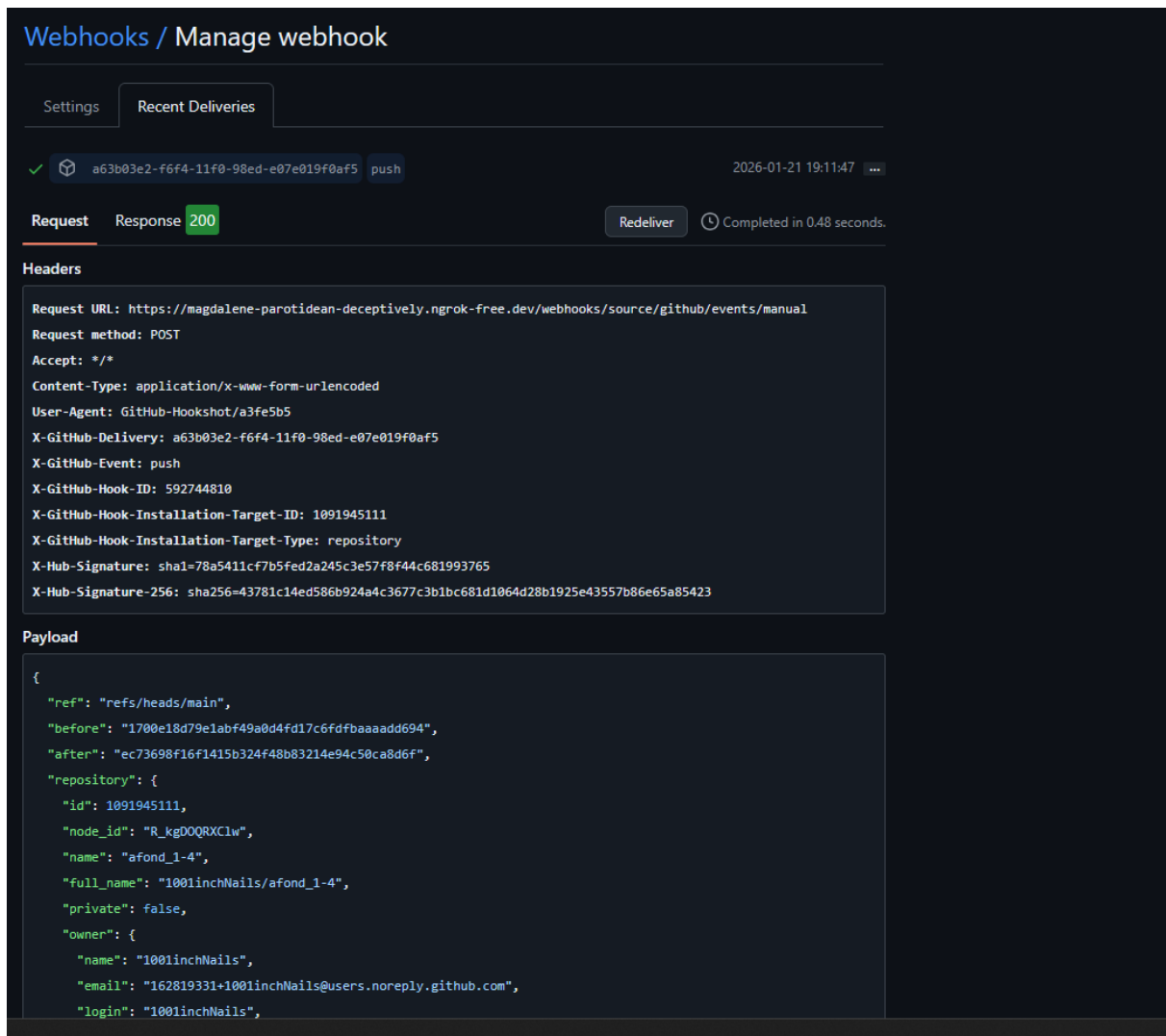


Figura 46: Success!

Volvemos al navegador y F5 para comprobar que el auto deploy se ha ejecutado correctamente

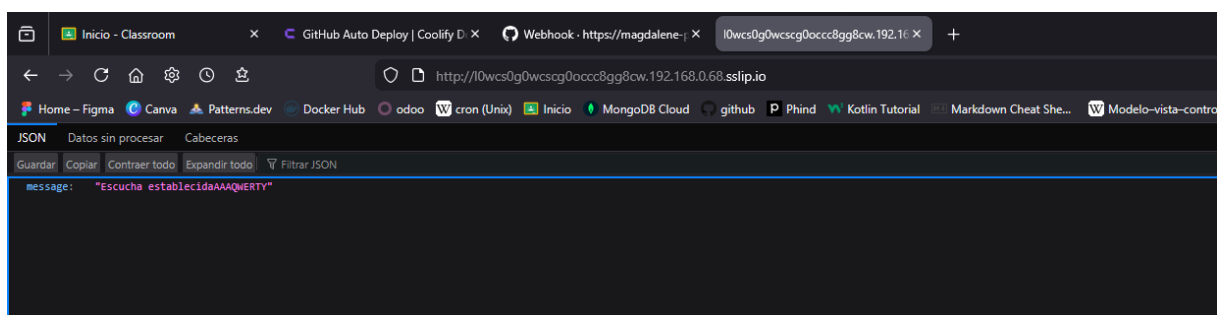


Figura 47: Oh yeah!

7 Info

Volver

Repositorio

Documentacion https://github.com/1001inchNails/afond_1_5

API de prueba https://github.com/1001inchNails/afond_1-4

Comandos:

- Instalar Coolify: `curl -fsSL https://cdn.coollabs.io/coolify/install.sh sudo bash|`

- Instalar Ngrok:

```
curl -sSL https://ngrok-agent.s3.amazonaws.com/ngrok.asc
sudo tee /etc/apt/trusted.gpg.d/ngrok.asc >/dev/null
&& echo "deb https://ngrok-agent.s3.amazonaws.com bookworm main"
| sudo tee /etc/apt/sources.list.d/ngrok.list
&& sudo apt update
&& sudo apt install ngrok
```

- Añadir authtoken: `ngrok config add-authtoken <authtoken>`
- Información contenedores Docker: `docker ps`
- IP de contenedor: `docker inspect -f '{{range .NetworkSettings.Networks}}{{.IPAddress}}`
- Exponer API test con Ngrok: `ngrok http <ip>:<puerto>`
- Exponer puerto con Ngrok: `ngrok http <puerto>`
- Interfaces de red: `ifconfig`